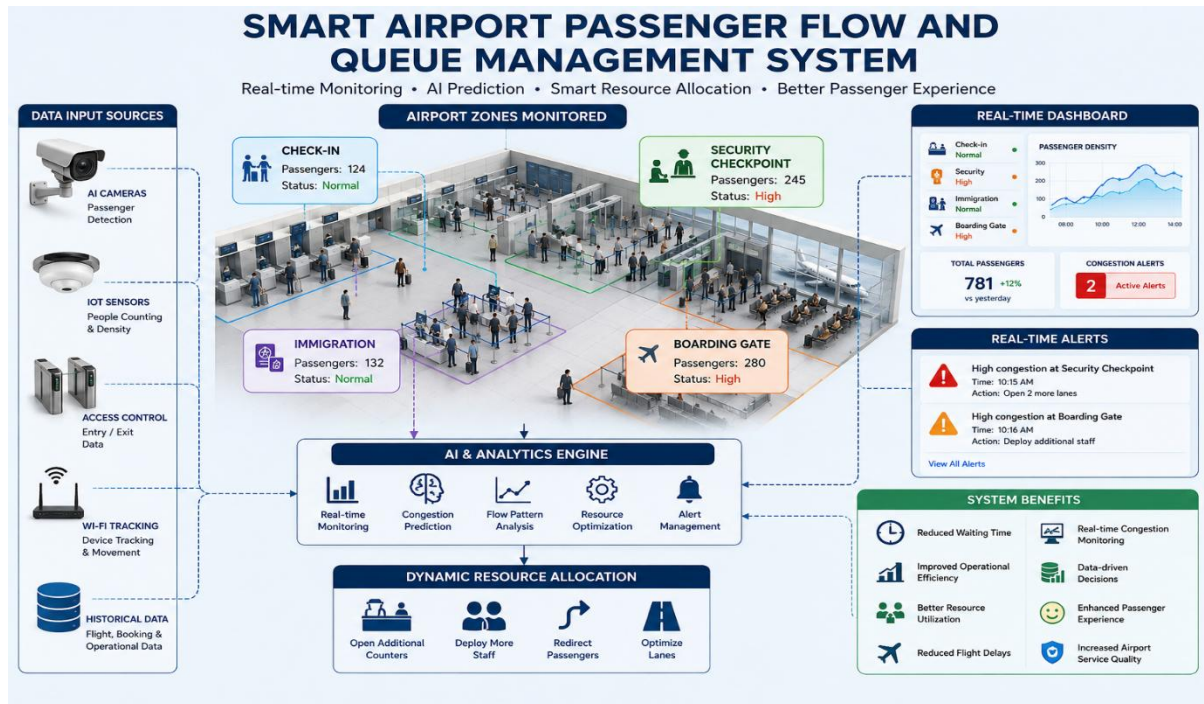


SMART AIRPORT PASSENGER FLOW AND QUEUE MANAGEMENT SYSTEM



NAME :M.HANIF RIYAZ

REG NO :192521123

CODE :CSA1013

1. PROBLEM STATEMENT

Airports handle millions of passengers annually, making efficient passenger movement essential for operational excellence and customer satisfaction. Long queues at check-in counters, security checkpoints, immigration counters, and boarding gates can lead to delays, increased waiting time, and poor passenger experience.

The Smart Airport Passenger Flow and Queue Management System is designed to continuously monitor passenger movement using AI-based cameras and IoT sensors. The system analyzes passenger density and queue information in real time, predicts possible congestion using AI algorithms, recommends optimal resource allocation, and generates real-time alerts for airport authorities.

The main purpose of the system is to reduce passenger waiting time, improve airport operational efficiency, optimize resource utilization, reduce flight delays, and enhance passenger satisfaction.

2. OBJECTIVES

1. To monitor passenger movement in real time.
2. To predict passenger congestion using AI models.
3. To optimize queue management.

4. To allocate airport resources dynamically.
 5. To generate real-time congestion alerts.
 6. To analyze passenger flow patterns.
 7. To improve security checkpoint efficiency.
 8. To enhance passenger satisfaction.
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3. FUNCTIONAL REQUIREMENTS

FR1 – Passenger Flow Monitoring

The system shall monitor passenger movement in different airport areas in real time.

FR2 – Passenger Density Detection

The system shall detect and calculate passenger density in different airport zones.

FR3 – Queue Monitoring

The system shall monitor queues at check-in counters, security checkpoints, immigration counters, and boarding gates.

FR4 – Congestion Prediction

The system shall use AI algorithms to predict possible passenger congestion.

FR5 – Dynamic Resource Allocation

The system shall recommend or allocate additional counters, security lanes, or staff when congestion is detected.

FR6 – Real-Time Alerts

The system shall generate alerts when passenger density or queue length exceeds the predefined threshold.

FR7 – Passenger Flow Analysis

The system shall analyze historical passenger-flow patterns.

FR8 – Security Checkpoint Monitoring

The system shall monitor security checkpoint efficiency and waiting time.

FR9 – Real-Time Dashboard

The system shall provide airport authorities with a dashboard showing passenger density, queue status, alerts, and resource availability.

FR10 – Data Storage

The system shall store passenger-flow and queue information for future analysis and reporting.

4. NON-FUNCTIONAL REQUIREMENTS

NFR1 – Performance

The system should process passenger-flow information with minimum delay.

NFR2 – Reliability

The system should operate continuously and provide reliable monitoring.

NFR3 – Scalability

The system should support increasing passenger volume and additional airport zones.

NFR4 – Security

Passenger-related data and system information should be protected from unauthorized access.

NFR5 – Availability

The system should remain available during airport operating hours.

NFR6 – Usability

The dashboard should be simple and easy for airport authorities to understand.

NFR7 – Accuracy

Passenger density detection and congestion prediction should provide accurate results.

NFR8 – Maintainability

The system should allow easy maintenance and updating of monitoring components and AI models.

5. TEST SCENARIOS

Scenario ID Test Scenario

TS01	Verify passenger movement monitoring
TS02	Verify passenger density detection
TS03	Verify queue length monitoring
TS04	Verify congestion prediction
TS05	Verify congestion alert generation
TS06	Verify dynamic resource allocation
TS07	Verify security checkpoint monitoring
TS08	Verify real-time dashboard updates
TS09	Verify historical passenger-flow analysis
TS10	Verify IoT sensor data reception
TS11	Verify camera-based passenger detection
TS12	Verify low passenger-density condition
TS13	Verify threshold passenger-density condition

Scenario ID Test Scenario

TS14	Verify high passenger-density condition
TS15	Verify camera or sensor failure
TS16	Verify unauthorized dashboard access
TS17	Verify system performance during peak hours
TS18	Verify recovery after network failure

6. DETAILED TEST CASES

TC ID Test Scenario		Actual Result Status	
TC01	Passenger movement monitoring	To be filled	Pass/Fail
TC02	Passenger density detection	To be filled	Pass/Fail
TC03	Queue monitoring	To be filled	Pass/Fail
TC04	Low passenger density	To be filled	Pass/Fail
TC05	Threshold condition	To be filled	Pass/Fail
TC06	High passenger density	To be filled	Pass/Fail
TC07	Congestion alert	To be filled	Pass/Fail
TC08	Congestion prediction	To be filled	Pass/Fail
TC09	Resource allocation	To be filled	Pass/Fail
TC10	Dashboard update	To be filled	Pass/Fail
TC11	Camera monitoring	To be filled	Pass/Fail
TC12	Sensor failure	To be filled	Pass/Fail
TC13	Network failure	To be filled	Pass/Fail

TC ID	Test Scenario	Actual Result Status	
TC14	Unauthorized login	To be filled	Pass/Fail
TC15	Authorized login	To be filled	Pass/Fail
TC16	Historical analysis	To be filled	Pass/Fail
TC17	Peak-hour monitoring	To be filled	Pass/Fail
TC18	Security checkpoint efficiency	To be filled	Pass/Fail
TC19	Invalid sensor data	To be filled	Pass/Fail
TC20	Alert clearance	To be filled	Pass/Fail

7. BOUNDARY VALUE ANALYSIS

Boundary Value Analysis is used to test passenger-density and queue-length limits.

Assume the maximum acceptable queue length is 100 passengers.

Test Case	Input Value	Condition	Expected Result
BVA01	98	Below boundary	Normal status
BVA02	99	Just below boundary	Normal status

Test Case	Input Value	Condition	Expected Result
BVA03	100	Boundary value	Threshold condition detected
BVA04	101	Just above boundary	Congestion warning generated
BVA05	102	Above boundary	Congestion alert generated

Boundary Value Analysis ensures that the system behaves correctly at and around the congestion threshold.

8. EQUIVALENCE PARTITIONING

Passenger density is divided into different input classes.

Partition ID	Passenger Count	Input Category	Expected System Behavior
EP01	0–50	Low Density	System displays Normal/Low Density status
EP02	51–100	Moderate Density	System displays Moderate Density status
EP03	101–200	High Density	System identifies High Congestion
EP04	Above 200	Severe Density	System generates Critical Congestion Alert
EP05	Negative Value	Invalid Input	System rejects the input and displays an error
EP06	Non-numeric Value	Invalid Input	System rejects the input and displays an error

Equivalence Partitioning reduces the number of test cases while ensuring that different categories of input are tested.

9. DECISION TABLE TESTING

Decision Table Testing is used to determine whether congestion alerts and additional resources are required.

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
Passenger Density High?	No	Yes	Yes	Yes
Queue Length High?	No	No	Yes	Yes
AI Predicts Congestion?	No	Yes	Yes	No
Generate Alert	No	Yes	Yes	Yes
Allocate Additional Resources	No	Yes	Yes	No
Update Dashboard	Yes	Yes	Yes	Yes

Rule Explanation

Rule 1: Passenger density and queue length are normal. No congestion alert is required.

Rule 2: AI predicts congestion. The system generates an alert and recommends additional resources.

Rule 3: Passenger density and queue length are high. The system generates an alert and recommends additional resources.

Rule 4: High passenger density or queue condition requires an alert even when AI prediction is unavailable.

10. STATE TRANSITION TESTING

The system can operate in the following states:

1. Normal

2. Warning
3. Congestion
4. Critical Congestion
5. Recovery

Test Case	Current State	Event/Condition	Next State	Expected Result
ST01	Normal	Passenger density increases	Warning	Warning status is displayed
ST02	Warning	Passenger density continues increasing	Congestion	Congestion alert is generated
ST03	Congestion	Passenger density becomes very high	Critical Congestion	High-priority alert is generated
ST04	Critical Congestion	Additional resources allocated	Congestion	Resource status is updated
ST05	Congestion	Passenger density decreases	Recovery	Recovery status is displayed
ST06	Recovery	Passenger density returns to normal	Normal	Congestion alert is cleared

11. EXCEPTIONAL TEST CASES

Test Case ID	Exceptional Condition	Expected Result
EX01	Camera stops working	Camera failure notification is displayed
EX02	IoT sensor disconnects	Missing sensor data is detected
EX03	Network connection is lost	Communication failure notification is displayed
EX04	Invalid passenger count is received	Invalid data is rejected
EX05	AI prediction service becomes unavailable	Basic monitoring continues and warning is displayed
EX06	Database becomes unavailable	Database error is displayed and recovery is handled
EX07	Unauthorized dashboard access is attempted	Access is denied
EX08	Extremely high passenger volume occurs	System remains stable and generates a critical alert

12. NON-FUNCTIONAL TEST CASES

Test Case ID	Requirement	Test Condition	Expected Result
NFT01	Performance	Monitor a large passenger volume	System responds within an acceptable time
NFT02	Reliability	Run the system continuously	System operates without unexpected failure

Test Case ID	Requirement	Test Condition	Expected Result
NFT03	Security	Attempt unauthorized login	Access is denied
NFT04	Usability	Airport staff use the dashboard	Information is clear and easy to understand
NFT05	Scalability	Add additional airport zones	System supports additional zones
NFT06	Availability	Run the system during airport operating hours	System remains available
NFT07	Accuracy	Compare detected and actual passenger count	Passenger detection provides acceptable accuracy
NFT08	Recovery	Restore network after failure	System resumes monitoring correctly

13. REQUIREMENT TRACEABILITY MATRIX

Requirement ID	Requirement	Test Case IDs
FR1	Passenger Flow Monitoring	TC01, TC11
FR2	Passenger Density Detection	TC02, TC04, TC05, TC06
FR3	Queue Monitoring	TC03, TC18
FR4	Congestion Prediction	TC08
FR5	Dynamic Resource Allocation	TC09

Requirement ID	Requirement	Test Case IDs
FR6	Real-Time Alerts	TC07, TC20
FR7	Passenger Flow Analysis	TC16
FR8	Security Checkpoint Monitoring	TC18
FR9	Real-Time Dashboard	TC10, TC15
FR10	Data Storage	TC16
NFR1	Performance	NFT01
NFR2	Reliability	NFT02
NFR3	Scalability	NFT05
NFR4	Security	NFT03
NFR5	Availability	NFT06
NFR6	Usability	NFT04
NFR7	Accuracy	NFT07
NFR8	Recovery/Maintainability	NFT08

14. TEST COVERAGE

The designed test cases provide coverage for the following areas:

1. Functional requirements
2. Non-functional requirements
3. Normal operating conditions
4. Boundary conditions
5. Invalid input conditions

6. Exceptional conditions
7. Passenger-density monitoring
8. Queue management
9. AI-based congestion prediction
10. Dynamic resource allocation
11. Real-time congestion alerts
12. Security checkpoint monitoring
13. Real-time dashboard functionality
14. Camera and IoT sensor failures
15. Network failures
16. Authentication and security
17. Performance and scalability
18. Recovery after system failures

The test cases provide adequate coverage of the major functional and quality requirements of the Smart Airport Passenger Flow and Queue Management System.

15. EXPECTED OUTCOMES

1. Reduced passenger waiting time.
2. Improved airport operational efficiency.
3. Better utilization of airport resources.
4. Enhanced passenger experience.
5. Reduced flight delays.
6. Real-time congestion monitoring.
7. Data-driven operational decisions.

8. Increased airport service quality.

16. CONCLUSION

The Smart Airport Passenger Flow and Queue Management System requires comprehensive testing to ensure reliable passenger monitoring, accurate congestion prediction, efficient queue management, and timely resource allocation.

The proposed test cases cover normal, boundary, invalid, and exceptional conditions. Testing techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing are applied to improve test coverage and system reliability.

Successful implementation and testing of this system can reduce passenger waiting time, improve airport resource utilization, minimize congestion and delays, and enhance the overall passenger experience.